

# TRANSFERABLE ADVERSARIAL ROBUSTNESS FOR SPEECH FOUNDATION MODELS VIA HIERARCHICAL STABILIZATION

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## ABSTRACT

Frozen speech foundation models (SFM) make downstream adaptation efficient: the backbone can stay fixed while a task learns layer fusion and a lightweight classifier. Full adversarial fine-tuning is a standard route to robustness, but generating adversarial examples and updating the backbone for every task sacrifices that efficiency. We ask whether robustness can instead be learned before future tasks are known. For a frozen backbone and linear classifier, robustness can be understood through the interaction between representation stability and decision-boundary margin. This leads directly to our design: we stabilize representations across the hidden layers, rather than only the final layer, while preserving clean representations; after clean adaptation selects the layer mixture, we keep it fixed and enlarge only the classifier margin, without downstream adversarial examples. We evaluate Wav2Vec2, HuBERT, and WavLM Large on four tasks under adaptive 30 dB attacks. Across 12 backbone-task pairs, hierarchical robustification improves robust accuracy by 46.4 pp, while margin refinement adds 4.0 pp for 1.1 pp of clean accuracy. Code and configurations are available at [github.com/arefmousavi/hierarchical-robust-sfm](https://github.com/arefmousavi/hierarchical-robust-sfm).

**Index Terms**— speech foundation models, adversarial robustness, adversarial training, transfer learning, layer fusion

## 1. INTRODUCTION

Frozen speech foundation models (SFM) support efficient downstream adaptation because the backbone can remain fixed. In the SUPERB-style classification setting considered here, a task learns a weighted fusion of frozen layer representations together with a lightweight classifier [1]. Large-scale evaluation shows that learnable layer fusion outperforms final-layer-only evaluation in most cases, while preferred depths vary across tasks [2, 3]. Full task-specific adversarial fine-tuning is a direct route to robustness, but iterative adversarial-example generation and end-to-end backbone updates are costly and must be repeated for every new task [4, 5]. We therefore ask whether the efficiency of frozen-backbone adaptation can be retained while robustness is learned before future downstream tasks are known.

This goal requires more than a robust terminal embedding. FARE in Robust CLIP stabilizes the CLIP vision-encoder output so existing downstream systems can reuse the robust encoder without retraining [6, 7]. In our setting, however, the future speech task chooses both a layer mixture and a classifier after backbone robustification. For a frozen backbone and linear classifier, robustness is governed by the interaction between representation stability and decision-boundary margin. Let  $z_\alpha(x)$  be the fused representation,

$r_\alpha(x, \delta) = z_\alpha(x + \delta) - z_\alpha(x)$  the attack-induced movement, and  $m(x)$  the Euclidean distance to the nearest classifier decision boundary. A sufficient condition for preserving the prediction is

$$\|r_\alpha(x, \delta)\|_2 < m(x). \quad (1)$$

This view leads directly to our design. Before tasks are known, we stabilize representations across the hidden layers, rather than only the final layer, while preserving clean representations. After clean adaptation selects the task-specific fusion and classifier, we keep the backbone and fusion fixed and refine only the classifier margin. Keeping the fusion fixed lets the margin objective change decision geometry without changing the representation selected by clean adaptation, and requires no downstream adversarial examples. Fig. 1 summarizes the resulting pipeline and its movement-margin interpretation.

Prior work establishes that robust pretrained representations can transfer robustness to downstream classifiers and studies how that robustness behaves during adaptation [5, 8, 9, 10, 11]. In speech, Wu et al. study attacks when the downstream task, layer-fusion weights, and classifier are unknown to the attacker [12]. Olivier et al. demonstrate adversarial transfer across self-supervised speech models [13], and Wang et al. study task-specific adversarial fine-tuning for robust ASR [14]. These works do not address the combination studied here: future task-specific layer fusion and classifier are unknown during robustification, representations must therefore be stabilized across layers, and the downstream decision geometry is refined after adaptation, using no adversarial examples.

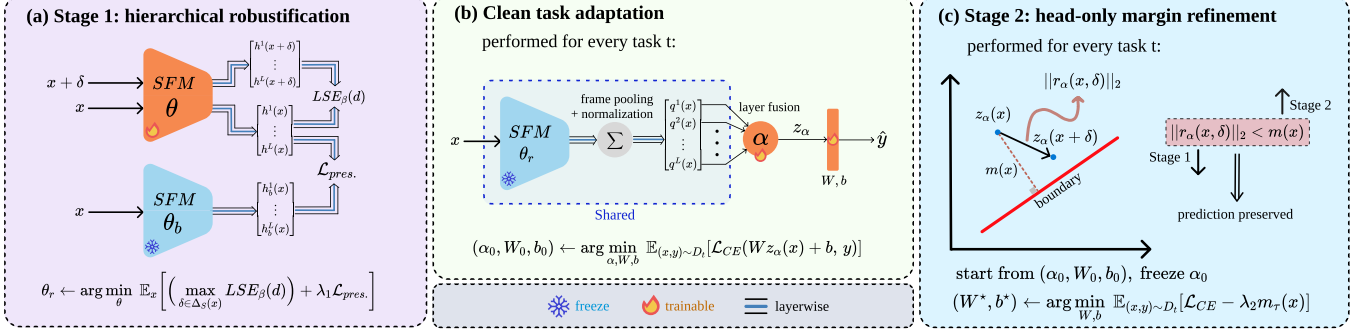
Our main contributions are: **(1) Hierarchy-aware adversarial robustification.** We introduce an adversarial objective matched to frozen SFM layer fusion that stabilizes representations across hidden layers while preserving clean representations. **(2) Movement-margin formulation.** We characterize robustness through the interaction between adversarial representation movement and distance to the nearest classifier boundary, and show that across-layer stabilization bounds this movement for every layer fusion a task may select. **(3) Clean margin-aware classifier refinement.** We refine only the downstream classifier, with the robust backbone and learned fusion fixed, using no downstream adversarial examples. We evaluate Wav2Vec2, HuBERT, and WavLM Large on four complementary speech classification tasks using adaptive white-box attacks.

## 2. METHOD

### 2.1. Notation and layer-normalized representations

Let  $x$  be a waveform and let  $h_\theta^\ell(x) \in \mathbb{R}^{T_x \times D}$  be the sequence of  $T_x$  valid (unpadded) frames that a backbone with parameters  $\theta$  produces at layer  $\ell$ , where  $D$  is the hidden dimension and  $\ell = 1, \dots, L$  indexes the outputs of the  $L$  transformer blocks. All  $L$  representations

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**Fig. 1.** Overview of the proposed framework. **STAGE 1** robustifies representations across hidden layers before downstream tasks are known, while preserving clean representations. Clean adaptation then learns task-specific convex layer fusion and a linear classifier with the robust backbone frozen. **STAGE 2** fixes the backbone and fusion and refines only the classifier using normalized geometric margins.

of an utterance share the frame count  $T_x$ . The pretrained backbone is  $\theta_b$  and its robustified counterpart  $\theta_r$ ;  $\Pi(h) = \frac{1}{T_x} \sum_{t=1}^{T_x} h_t \in \mathbb{R}^D$  denotes mean pooling over valid frames.

Layers cannot be compared on raw distances, because activation scale varies across depth. We therefore fix one scale per layer,  $\sigma_\ell^2 = \mathbb{E}_x \left[ \frac{1}{T_x D} \|h_{\theta_b}^\ell(x)\|_F^2 \right]$ , estimated once on held-out clean speech independent of any downstream task, and measure the discrepancy between two frame sequences  $a, b \in \mathbb{R}^{T_x \times D}$  by

$$\rho_\ell^2(a, b) = \frac{1}{T_x D \sigma_\ell^2} \sum_{t=1}^{T_x} \|a_t - b_t\|_2^2. \quad (2)$$

Because the  $\sigma_\ell$  are estimated from  $\theta_b$  and then held fixed, no method can shrink Eq. (2) by inflating its own normalizer, so layer drift stays comparable across methods. A downstream task consumes the same layers after pooling, centering, and normalization:

$$q_\ell(x) = \frac{\Pi(h_{\theta_r}^\ell(x)) - \mu_\ell}{\sigma_\ell \sqrt{D}} \in \mathbb{R}^D, \quad z_\alpha(x) = \sum_{\ell=1}^L \alpha_\ell q_\ell(x), \quad (3)$$

where  $\mu_\ell$  is the mean of  $\Pi(h_{\theta_r}^\ell(x))$  over a held-out split of the task data  $D_t$ , and the fusion weights satisfy  $\alpha_\ell \geq 0$  and  $\sum_{\ell=1}^L \alpha_\ell = 1$ . A linear classifier  $g(z) = Wz + b$ , with  $W \in \mathbb{R}^{K \times D}$  and rows  $w_1^\top, \dots, w_K^\top$ , acts on  $z_\alpha$ . Because Eq. (2) and Eq. (3) share  $\sigma_\ell$ , robustification and downstream adaptation are expressed in one coordinate system.

## 2.2. STAGE 1: adversarial stability across layers

At robustification time the future task, its fusion weights, and its classifier are all unknown, so no layer can be favored in advance. We score the instability of layer  $\ell$  by  $d_\ell(x, \delta; \theta) = \rho_\ell^2(h_{\theta_b}^\ell(x), h_{\theta_b}^\ell(x + \delta))$  and aggregate  $\mathbf{d} = (d_1, \dots, d_L)$  across depth with the smooth maximum

$$LSE_\beta(\mathbf{d}) = \frac{1}{\beta} \log \left[ \frac{1}{L} \sum_{\ell=1}^L e^{\beta d_\ell} \right], \quad (4)$$

which brackets the worst layer:  $\max_\ell d_\ell - \frac{1}{\beta} \log L \leq LSE_\beta(\mathbf{d}) \leq \max_\ell d_\ell$ , with  $\beta \rightarrow 0$  recovering the layer mean. Robustness must not destroy the pretrained features, and  $\mathbf{d}$  can also be driven down by shrinking activations, so we anchor the clean representation to  $\theta_b$ ,

$$\mathcal{L}_{pres.}(x) = \frac{1}{L} \sum_{\ell=1}^L \rho_\ell^2(h_{\theta_b}^\ell(x), h_{\theta_b}^\ell(x)), \quad (5)$$

and, using unlabeled task-independent speech, solve

$$\theta_r = \arg \min_{\theta} \mathbb{E}_x \left[ \left( \max_{\delta \in \Delta_S(x)} LSE_\beta(\mathbf{d}) \right) + \lambda_1 \mathcal{L}_{pres.}(x) \right], \quad (6)$$

where  $\Delta_S(x)$  is the perturbation set of Eq. (12) at  $S = 30$  dB and the inner maximization is approximated by projected gradient ascent.  $\mathcal{L}_{pres.}$  sits outside it because it does not depend on  $\delta$ .

**Why the objective must span layers.** Let  $\Delta q_\ell = q_\ell(x + \delta) - q_\ell(x)$ , so that  $r_\alpha(x, \delta) = \sum_\ell \alpha_\ell \Delta q_\ell$ . Pooling averages over frames, so Jensen’s inequality gives  $\|\Delta q_\ell\|_2^2 \leq d_\ell$ , both sides at  $\theta_r$ : a layer’s pooled displacement cannot exceed the frame-level quantity STAGE 1 minimizes. With the convexity of the mixture and the two-sided bound on  $LSE_\beta$ ,

$$\begin{aligned} \|r_\alpha(x, \delta)\|_2 &\leq \max_\ell \|\Delta q_\ell\|_2 \leq (\max_\ell d_\ell)^{1/2} \\ &\leq (LSE_\beta(\mathbf{d}) + \frac{1}{\beta} \log L)^{1/2} \end{aligned} \quad (7)$$

for every admissible  $\alpha$ : STAGE 1 controls the movement term of Eq. (1) uniformly over the fusions a future task might choose. Minimizing  $d_L$  alone leaves the other  $d_\ell$  unconstrained, so it bounds  $\|r_\alpha\|_2$  only for the mixture that uses layer  $L$  by itself.

## 2.3. Clean adaptation and the movement-margin condition

For task  $t$  we freeze  $\theta_r$  and learn the fusion weights and linear classifier from clean labels alone,

$$(\alpha_0, W_0, b_0) = \arg \min_{\alpha, W, b} \mathbb{E}_{(x,y) \sim D_t} \mathcal{L}_{CE}(W z_\alpha(x) + b, y), \quad (8)$$

which fixes the task representation before any margin refinement. For a true class  $y$  and a competing class  $j \neq y$ , the signed Euclidean distance from  $z_{\alpha_0}(x)$  to the boundary  $\{z : (w_y - w_j)^\top z + b_y - b_j = 0\}$  is

$$m_{yj}(x) = \frac{(w_y - w_j)^\top z_{\alpha_0}(x) + (b_y - b_j)}{\|w_y - w_j\|_2}. \quad (9)$$

The decision region of class  $y$  is an intersection of  $K - 1$  half-spaces, so for correctly classified  $x$  the nearest-boundary margin  $m(x) = \min_{j \neq y} m_{yj}(x)$  is exactly the distance from  $z_{\alpha_0}(x)$  to the boundary of that region. Each  $m_{yj}$  is 1-Lipschitz in  $z$ , so a perturbation can shrink it by at most  $\|r\|_2$ , where  $r = r_{\alpha_0}(x, \delta)$ :  $m_{yj}(x + \delta) \geq m_{yj}(x) - \|r\|_2$ . If  $\|r\|_2 < m(x)$ , every margin stays positive and the prediction does not change, which is Eq. (1). Robustness therefore splits into two independently measurable quantities: movement, bounded by STAGE 1 through Eq. (7), and margin, enlarged by STAGE 2. The condition is sufficient, not necessary, so

**Table 1.** Robustness transfer across three backbones and four tasks. Task entries are clean/robust accuracy (CA/RA, %), reported as mean $\pm$ SD over three downstream adaptation seeds. The four overbarred columns report changes relative to Standard+CE, averaged across tasks within each backbone;  $\Delta CA$  and  $\Delta RA$  are in percentage points, while  $\Delta R_\alpha$  and  $\Delta M_{10}$  are defined in Eq. (13).  $\dagger$  indicates use of downstream adversarial examples; Full AT is a task-specific oracle outside the frozen-backbone setting.

| Method                                            | Clean / robust accuracy (%) $\uparrow$ |                                   |                                   |                                   | $\Delta R_\alpha$ | $\Delta M_{10}$                 | $\Delta CA$     | $\Delta RA$     |
|---------------------------------------------------|----------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-------------------|---------------------------------|-----------------|-----------------|
|                                                   | KS                                     | IC                                | SID                               | ER                                | (%) $\downarrow$  | ( $\times 10^{-3}$ ) $\uparrow$ | (pp) $\uparrow$ | (pp) $\uparrow$ |
| <b>Wav2Vec2 Large (primary backbone)</b>          |                                        |                                   |                                   |                                   |                   |                                 |                 |                 |
| Standard + CE                                     | 95.2 $\pm$ 0.56 / 5.5 $\pm$ 0.96       | 94.1 $\pm$ 0.47 / 2.7 $\pm$ 0.52  | 87.6 $\pm$ 0.91 / 0.5 $\pm$ 0.09  | 67.5 $\pm$ 0.52 / 0.8 $\pm$ 0.18  | 0.0               | 0.0                             | 0.0             | 0.0             |
| FARE-style + CE                                   | 91.7 $\pm$ 0.19 / 39.7 $\pm$ 0.71      | 90.5 $\pm$ 0.56 / 36.8 $\pm$ 0.63 | 83.9 $\pm$ 0.48 / 16.0 $\pm$ 0.99 | 64.9 $\pm$ 1.48 / 9.7 $\pm$ 0.95  | -38.3             | 0.5                             | -3.4            | 23.2            |
| Terminal-layer + CE                               | 92.5 $\pm$ 0.54 / 38.2 $\pm$ 0.75      | 91.2 $\pm$ 0.17 / 36.1 $\pm$ 1.36 | 84.1 $\pm$ 0.92 / 15.6 $\pm$ 0.58 | 64.8 $\pm$ 0.62 / 9.8 $\pm$ 1.43  | -37.5             | 2.1                             | -3.0            | 22.6            |
| STAGE 1 + CE                                      | 93.7 $\pm$ 0.56 / 62.2 $\pm$ 0.47      | 92.8 $\pm$ 0.58 / 61.0 $\pm$ 0.45 | 84.4 $\pm$ 0.47 / 34.8 $\pm$ 0.64 | 65.2 $\pm$ 0.73 / 26.5 $\pm$ 1.27 | -81.8             | 8.6                             | -2.1            | 43.8            |
| + STAGE 2 (ours)                                  | 92.3 $\pm$ 0.35 / 67.4 $\pm$ 1.22      | 91.6 $\pm$ 0.55 / 66.7 $\pm$ 1.15 | 83.6 $\pm$ 0.49 / 37.7 $\pm$ 0.65 | 64.0 $\pm$ 0.38 / 29.6 $\pm$ 0.70 | -82.2             | 13.1                            | -3.2            | 48.0            |
| + adversarial head training $\dagger$             | 90.4 $\pm$ 0.57 / 71.8 $\pm$ 0.53      | 89.4 $\pm$ 0.24 / 69.0 $\pm$ 0.54 | 81.5 $\pm$ 0.68 / 39.8 $\pm$ 0.97 | 62.1 $\pm$ 0.31 / 31.8 $\pm$ 0.60 | -82.6             | 14.0                            | -5.3            | 50.7            |
| <b>HuBERT Large</b>                               |                                        |                                   |                                   |                                   |                   |                                 |                 |                 |
| Standard + CE                                     | 94.2 $\pm$ 0.54 / 5.9 $\pm$ 1.04       | 97.0 $\pm$ 0.56 / 4.6 $\pm$ 0.57  | 90.6 $\pm$ 0.58 / 0.7 $\pm$ 0.13  | 68.9 $\pm$ 1.12 / 0.9 $\pm$ 0.17  | 0.0               | 0.0                             | 0.0             | 0.0             |
| FARE-style + CE                                   | 91.5 $\pm$ 0.63 / 40.6 $\pm$ 0.89      | 93.7 $\pm$ 0.61 / 43.7 $\pm$ 0.30 | 87.3 $\pm$ 1.35 / 17.9 $\pm$ 0.70 | 66.2 $\pm$ 0.84 / 11.3 $\pm$ 0.60 | -40.5             | 3.5                             | -3.0            | 25.4            |
| Terminal-layer + CE                               | 90.7 $\pm$ 0.32 / 42.1 $\pm$ 0.73      | 94.4 $\pm$ 0.53 / 42.3 $\pm$ 0.73 | 87.0 $\pm$ 0.50 / 18.6 $\pm$ 0.35 | 66.3 $\pm$ 0.39 / 11.4 $\pm$ 1.63 | -41.3             | 3.4                             | -3.1            | 25.6            |
| STAGE 1 + CE                                      | 92.6 $\pm$ 0.49 / 64.3 $\pm$ 1.24      | 95.9 $\pm$ 0.58 / 67.9 $\pm$ 0.54 | 87.5 $\pm$ 1.22 / 37.1 $\pm$ 1.08 | 66.8 $\pm$ 0.81 / 29.1 $\pm$ 0.63 | -85.9             | 9.2                             | -2.0            | 46.6            |
| + STAGE 2 (ours)                                  | 91.2 $\pm$ 0.54 / 69.4 $\pm$ 1.29      | 94.5 $\pm$ 0.53 / 72.5 $\pm$ 1.36 | 86.8 $\pm$ 0.53 / 41.5 $\pm$ 0.78 | 65.5 $\pm$ 0.33 / 32.7 $\pm$ 0.91 | -86.1             | 13.8                            | -3.2            | 51.0            |
| + adversarial head training $\dagger$             | 89.4 $\pm$ 0.42 / 72.8 $\pm$ 1.32      | 92.4 $\pm$ 0.27 / 74.2 $\pm$ 0.53 | 84.7 $\pm$ 0.81 / 43.3 $\pm$ 1.63 | 63.6 $\pm$ 0.61 / 34.9 $\pm$ 1.18 | -86.0             | 15.1                            | -5.2            | 53.3            |
| <b>WavLM Large</b>                                |                                        |                                   |                                   |                                   |                   |                                 |                 |                 |
| Standard + CE                                     | 96.1 $\pm$ 0.60 / 7.5 $\pm$ 1.41       | 96.7 $\pm$ 0.58 / 3.1 $\pm$ 0.09  | 94.3 $\pm$ 0.54 / 0.8 $\pm$ 0.15  | 71.2 $\pm$ 1.23 / 1.3 $\pm$ 0.47  | 0.0               | 0.0                             | 0.0             | 0.0             |
| FARE-style + CE                                   | 92.7 $\pm$ 0.54 / 44.0 $\pm$ 0.33      | 93.6 $\pm$ 0.55 / 40.4 $\pm$ 0.60 | 90.8 $\pm$ 0.39 / 20.8 $\pm$ 1.48 | 68.6 $\pm$ 0.40 / 12.8 $\pm$ 0.58 | -35.7             | 1.5                             | -3.2            | 26.3            |
| Terminal-layer + CE                               | 93.5 $\pm$ 0.33 / 42.8 $\pm$ 0.22      | 93.2 $\pm$ 0.47 / 41.9 $\pm$ 0.90 | 91.4 $\pm$ 0.75 / 19.3 $\pm$ 1.26 | 68.5 $\pm$ 1.02 / 12.9 $\pm$ 0.49 | -35.6             | 1.7                             | -2.9            | 26.1            |
| STAGE 1 + CE                                      | 94.4 $\pm$ 0.54 / 69.1 $\pm$ 1.20      | 95.3 $\pm$ 0.37 / 65.6 $\pm$ 0.44 | 91.3 $\pm$ 0.53 / 41.7 $\pm$ 0.76 | 68.9 $\pm$ 1.37 / 31.5 $\pm$ 1.12 | -80.5             | 5.6                             | -2.1            | 48.8            |
| + STAGE 2 (ours)                                  | 93.1 $\pm$ 0.20 / 71.6 $\pm$ 1.28      | 94.4 $\pm$ 0.58 / 71.1 $\pm$ 1.25 | 90.9 $\pm$ 1.27 / 44.4 $\pm$ 0.76 | 67.7 $\pm$ 0.22 / 34.0 $\pm$ 0.64 | -80.8             | 9.7                             | -3.1            | 52.1            |
| + adversarial head training $\dagger$             | 91.2 $\pm$ 0.29 / 75.2 $\pm$ 0.59      | 92.0 $\pm$ 0.34 / 73.5 $\pm$ 0.66 | 88.3 $\pm$ 1.06 / 48.0 $\pm$ 0.81 | 65.8 $\pm$ 0.77 / 36.9 $\pm$ 1.54 | -81.2             | 9.9                             | -5.3            | 55.2            |
| <b>Full downstream AT oracle – Wav2Vec2 Large</b> |                                        |                                   |                                   |                                   |                   |                                 |                 |                 |
| Standard + full AT $\dagger$                      | 91.0 $\pm$ 1.30 / 82.9 $\pm$ 3.09      | 90.8 $\pm$ 1.31 / 77.5 $\pm$ 3.05 | 82.1 $\pm$ 1.11 / 53.0 $\pm$ 1.97 | 62.6 $\pm$ 1.66 / 38.7 $\pm$ 2.49 | –                 | –                               | –               | –               |

we use the two as directional diagnostics rather than as a predictor of RA.

## 2.4. STAGE 2: clean classifier-margin refinement

Cross-entropy has no explicit control over the distance in Eq. (9): rescaling  $W$  and  $b$  raises every logit gap, and lowers the loss, without moving a boundary. STAGE 2 therefore optimizes that distance directly, and the normalization by  $\|w_y - w_j\|_2$  makes it immune to the same rescaling. STAGE 2 starts from  $(\alpha_0, W_0, b_0)$ , freezes  $\theta_r$  and  $\alpha_0$ , and updates only  $W$  and  $b$ . Freezing the fusion is deliberate: margin optimization reshapes decision geometry without reselecting what clean adaptation chose, so Eq. (7) still holds at  $\alpha_0$ . Because  $m(x)$  is a non-smooth minimum, we optimize a soft nearest-boundary margin

$$m_\tau(x) = -\tau \log \left[ \sum_{j \neq y} \exp \left( -\frac{m_{yj}(x)}{\tau} \right) \right], \quad (10)$$

which satisfies  $m(x) - \tau \log(K-1) \leq m_\tau(x) \leq m(x)$ : STAGE 2 maximizes a lower bound on the true margin, exact as  $\tau \rightarrow 0$ .

$$(W^*, b^*) = \arg \min_{W, b} \mathbb{E}_{(x, y) \sim \mathcal{D}_t} [\mathcal{L}_{CE} - \lambda_2 m_\tau(x)], \quad (11)$$

with  $\lambda_2 \geq 0$ . STAGE 2 never sees an adversarial example:  $\lambda_2, \tau$ , and the checkpoint come from clean training and validation data under a predeclared clean-accuracy constraint.

## 3. EXPERIMENTAL SETUP

**Backbones and tasks.** We evaluate Wav2Vec2 Large, HuBERT Large, and WavLM Large, which differ in pretraining objective and data [15, 16, 17]. We use Wav2Vec2 Large as the primary backbone for mechanism, attack, and sensitivity analyses, while all three backbones assess transfer breadth. We use four utterance-level tasks under the same frozen-backbone, learned-fusion, linear-head interface,

spanning complementary speech factors: Speech Commands keyword spotting (KS; content) [18], Fluent Speech Commands intent classification (IC; semantics) [19], VoxCeleb1 speaker identification (SID; speaker) [20], and four-class IEMOCAP emotion recognition (ER; paralinguistics) [21, 1]. Comparisons are within backbone, not across differently pretrained models [3].

**Threat model.** Adversarial audio spans direct waveform, psychoacoustic, and physical attacks [22, 23, 24, 25, 26]; we focus on untargated attacks with SNR-bounded digital waveform perturbations,

$$\Delta_S(x) = \left\{ \delta : \|\delta\|_2 \leq \|x\|_2 10^{-S/20}, x + \delta \in \mathcal{X} \right\}, \quad (12)$$

where  $\mathcal{X}$  is the admissible sample range and norms are taken over valid unpadded samples. Headline results use  $S = 30$  dB, with  $S \in \{24, 27, 30, 33, 36\}$  dB on the primary backbone for sensitivity analysis. The 30-dB operating point matches prior attacks on SSL speech recognizers [13]. Because global SNR is not a perceptual guarantee, we additionally include a psychoacoustically shaped speech-aware attack [24, 25].

**Robustness evaluation.** Following standard adversarial evaluation practice [27, 4, 28], we evaluate every model on the full test set using AutoPGD-CE [29], two defense-aware attacks that minimize the normalized classifier margin and maximize fused-representation displacement, respectively, and the psychoacoustically shaped attack. All four attacks run for 50 steps with 2 restarts, satisfy the 30 dB SNR constraint, and differentiate through preprocessing, pooling, learned fusion, and classifier. RA is the fraction of examples that remain correct under *every attack in the pool*, i.e., the per-example union rather than the minimum of separately averaged accuracies. To quantify the movement–margin mechanism, Table 1 reports each method’s changes relative to its matched Standard+CE baseline. Let  $\bar{R}_\alpha = \mathbb{E}_x \|r_\alpha(x, \delta_x)\|_2$  denote the mean fused-representation movement, where  $\delta_x$  is the attack in the pool that produces the largest displacement, computed against the model being measured. We de-

**Table 2.** Primary-backbone mechanism and validity suite. (a) STAGE 1 variants with clean CE adaptation and without STAGE 2. (b) CE, hard-margin, and soft-margin adaptation using the STAGE 1 backbone. (c) Attack-wise and per-example-union RA for our full STAGE 1+STAGE 2 method. All values are averaged over three downstream adaptation seeds; overbars then denote four-task means of changes relative to Standard+CE.

| (a) STAGE 1 variants (+CE)        |                            |                       |                       | (b) Margin refinement after STAGE 1 |                       |                                           |                       | (c) Full-method attack audit (RA) |      |      |      |      |
|-----------------------------------|----------------------------|-----------------------|-----------------------|-------------------------------------|-----------------------|-------------------------------------------|-----------------------|-----------------------------------|------|------|------|------|
| Variant                           | $\Delta R_\alpha$<br>(%) ↓ | $\Delta CA$<br>(pp) ↑ | $\Delta RA$<br>(pp) ↑ | Method                              | $\Delta RA$<br>(pp) ↑ | $\Delta M_{10}$<br>( $\times 10^{-3}$ ) ↑ | $\Delta CA$<br>(pp) ↑ | Attack                            | KS   | IC   | SID  | ER   |
| Terminal layer                    | -37.5                      | -3.0                  | 22.6                  | CE                                  | 43.8                  | 8.6                                       | -2.1                  | AutoPGD-CE                        | 69.5 | 73.1 | 40.5 | 34.7 |
| Mean all-layer                    | -72.9                      | -2.2                  | 39.3                  | hard margin                         | 45.5                  | 11.2                                      | -2.9                  | Normalized-margin (adaptive)      | 68.4 | 68.3 | 42.2 | 32.1 |
| STAGE 1, no $\mathcal{L}_{pres.}$ | -93.2                      | -20.8                 | 13.5                  | <b>soft margin (STAGE 2)</b>        | 48.0                  | 13.1                                      | -3.2                  | Fused-displacement (adaptive)     | 76.7 | 84.6 | 51.8 | 44.3 |
| <b>STAGE 1</b>                    | -81.8                      | -2.1                  | 43.8                  |                                     |                       |                                           |                       | Psychoacoustic                    | 79.5 | 86.3 | 53.0 | 45.8 |
|                                   |                            |                       |                       |                                     |                       |                                           |                       | <b>Per-example union</b>          | 67.4 | 66.7 | 37.7 | 29.6 |

fine

$$\Delta R_\alpha = \frac{\bar{R}_\alpha - \bar{R}_\alpha^{\text{Std+CE}}}{\bar{R}_\alpha^{\text{Std+CE}}}, \quad \Delta M_{10} = m_{10} - m_{10}^{\text{Std+CE}}, \quad (13)$$

where  $m_{10}$  is the 10th percentile of the clean normalized margin. Negative  $\Delta R_\alpha$  indicates reduced adversarial representation movement, whereas positive  $\Delta M_{10}$  indicates a larger lower-tail margin. On the primary backbone we verify the achieved SNR and confirm that RA no longer decreases beyond 50 steps and 2 restarts.

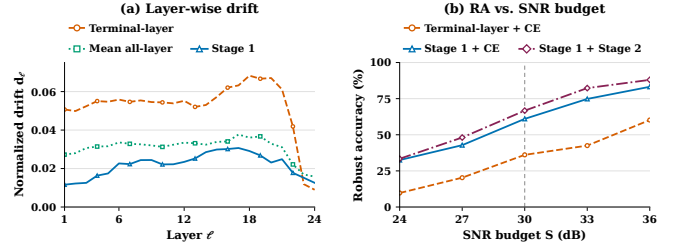
**Baselines and controls.** Table 1 compares Standard+CE, FARE-style robustification [7], a matched *Terminal-layer* control, our two-stage method, and downstream adversarial references. FARE-style minimizes one term, the displacement of the attacked final-layer output from its clean value under  $\theta_b$ , so robustness and preservation are not separated; it uses the same unlabeled data and 30-dB threat as STAGE 1. Terminal-layer instead keeps STAGE 1’s  $\mathcal{L}_{pres.}$  unchanged and replaces only  $\text{LSE}_\beta(d)$  by  $d_L$ . Adversarial head training keeps the STAGE 1 backbone and learned fusion fixed while training the classifier on downstream adversarial examples; full AT additionally updates the backbone and is reported only as a primary-backbone oracle.

**Training, selection, and reporting.** STAGE 1 uses 100h of Common Voice [30] as unlabeled robustification data and is trained for three epochs with  $\lambda_1 = 0.2$ , 10-step PGD at 30 dB, and  $\beta = 30$ , which causes  $\text{LSE}_\beta$  to emphasize higher-drift layers, especially at the start of robustification;  $\sigma_\ell$  is calibrated on a disjoint held-out split. STAGE 2 uses  $\lambda_2 = 2$ ,  $\tau = 0.1$ , and trains for 50 epochs on cached fused representations. Both stages use Adam with 10% linear warm-up and linear decay to zero, with peak learning rates  $5 \times 10^{-5}$  for STAGE 1 and  $10^{-3}$  for STAGE 2. STAGE 2 hyperparameters are selected on held-out clean data; no adversarial validation or test examples are used. Tables 1–2 report values across three downstream seeds for a fixed STAGE 1 checkpoint;  $\Delta$  quantities are paired by seed and overbars denote four-task means.

**Mechanism and validity audits.** On the primary backbone, Table 2(a) ablates STAGE 1’s across-depth aggregation, worst-layer weighting, and clean anchor. Panel (b) compares CE, hard-margin, and soft-margin objectives with the backbone and fusion fixed. Panel (c) reports attack-wise and per-example-union RA. Fig. 2 shows layer-wise drift and the 24–36 dB SNR sweep.

## 4. RESULTS AND ANALYSIS

**Robustness transfer.** STAGE 1+CE improves RA over Standard+CE on all 12 backbone–task pairs, by 46.4pp on average at a 2.1-pp CA cost, and exceeds the Terminal-layer control by 21.7pp RA at 0.9pp higher CA (Table 1). Stabilizing the hidden



**Fig. 2.** Primary-backbone diagnostics on the IC task. (a) Layer-wise representation drift under terminal-layer, mean all-layer, and STAGE 1 stabilization. (b) Robust accuracy across 24–36 dB SNR budgets.

hierarchy before the task is known therefore transfers robustness that terminal-layer stabilization does not.

**Clean margin refinement.** STAGE 2 adds RA on every pair, 4.0pp on average for 1.1 pp CA, using no downstream adversarial examples; the soft surrogate beats the hard minimum by 2.5 pp RA with a larger  $m_{10}$  (Table 2(b)). Using downstream adversarial examples, head training gains 2.7 pp more RA for 2.1 pp less CA; full AT reaches 63.0% mean RA against our 50.4% on the primary backbone, but updates the backbone per task.

**Hierarchy and preservation.** STAGE 1 exceeds mean all-layer stabilization by 4.5pp RA (Table 2(a)), and Fig. 2(a) shows why: the terminal-layer objective attains the lowest drift at layer  $L$  but the highest drift at most earlier depths, while worst-layer weighting pushes STAGE 1’s peak drift below the layer-mean variant. Removing  $\mathcal{L}_{pres.}$  gives the largest movement reduction (93.2%) yet costs 18.7 pp CA and 30.3 pp RA against STAGE 1: reduced movement alone is not robustness.

**Adaptive evaluation.** The normalized-margin attack is strongest on KS, IC, and ER and AutoPGD-CE on SID, and the union removes a further 1.0–2.8 pp, so the attacks fail on different examples (Table 2(c)). The method ranking is unchanged across 24–36 dB (Fig. 2(b)).

**Training cost.** On an RTX 4090, STAGE 1 costs 18.9 GPU-h once per backbone, while STAGE 2 costs 0.37 GPU-h per task, compared with 10.7 GPU-h per task for full AT.

## 5. CONCLUSION

Stabilizing the hidden hierarchy before tasks are known transfers robustness beyond terminal-layer stabilization; clean margin refinement further improves robustness without downstream adversarial examples. Across 12 backbone–task pairs, the full method improves mean RA by 50.4 pp over Standard+CE at a 3.2 pp CA cost under the evaluated digital waveform attacks.

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**Compliance with Ethical Standards.** This research was conducted retrospectively on existing human speech corpora (Common Voice, Speech Commands, Fluent Speech Commands, VoxCeleb1, IEMO-CAP), each obtained and used under its respective license.

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